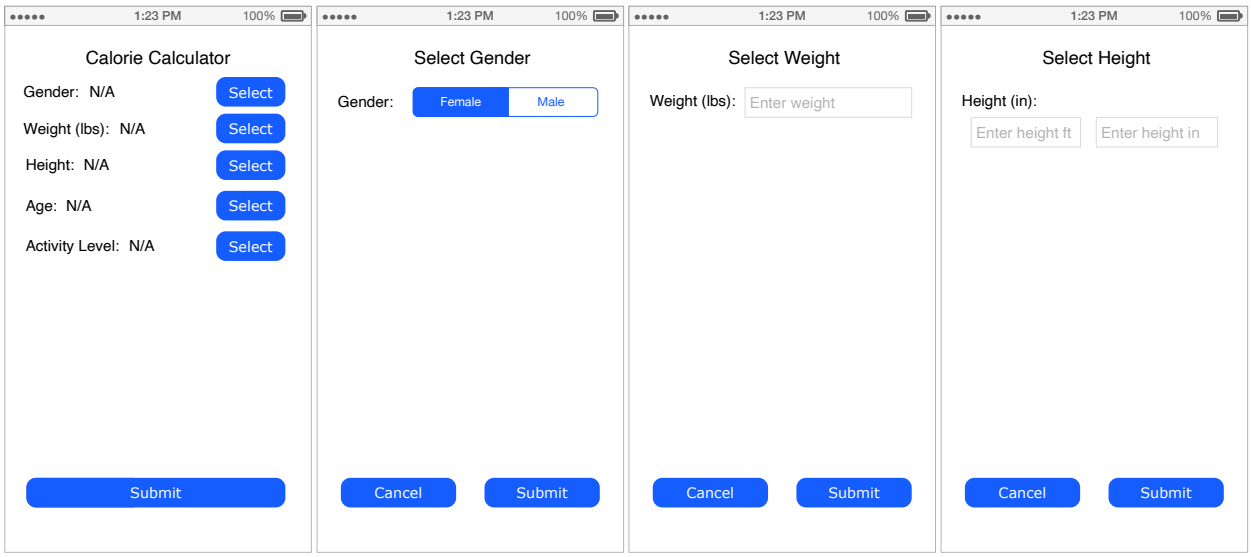
Mobile App Development
In-Class Assessment 1

Basic Instructions:

1. This is and In Class Assessment, which will count for 8% of the total course grade.
2. This assessment is an individual effort. Each student is responsible for her/his own assessment and its submission.
3. Once you have picked up the assessment, you may not discuss it in any way with anyone until the assessment period is over.
4. During the assessment, you are allowed to use the course videos, slides, and your code from previous home works and in class assignments. You can use the internet to search for answers. You are NOT allowed to use code provided by other students or solicit help from other online persons.
5. Answer all the assessment parts, all the parts are required.
6. During the assessment the teaching assistants and Instructors will pass by each student and ask them to demonstrate their application. Your interaction with the teaching assistants and instructors will be taken into consideration when grading your assessment submission.
7. Please download the support files provided with the assessment and use them when implementing your project.
8. Your assignment will be graded for functional requirements and efficiency of your submitted solution. You will loose points if your code is not efficient, does unnecessary processing or blocks the UI thread.
9. Create a zip file which includes all the project folder, any required libraries, and your presentation material. Submit the exported file using the provided canvas submission link.
10. **Do not try to use any Social Messenger apps, Emails, Or Cloud File Storage services in this exam.**
11. **Failure to follow the above instructions will result in point deductions.**
12. **Any violation of the rules regarding consultation with others will not be tolerated and will result disciplinary action and failing the course.**

In-Class Assessment 1 (100 Points)

In this assignment you will be building a daily calorie calculator app.



(a) Calorie Calculator **(b) Select Gender** **(c) Select Weight** **(d) Select Height**

Figure 1, Application User Interface

Part 1, Calorie Calculator UIViewController (20 Points):

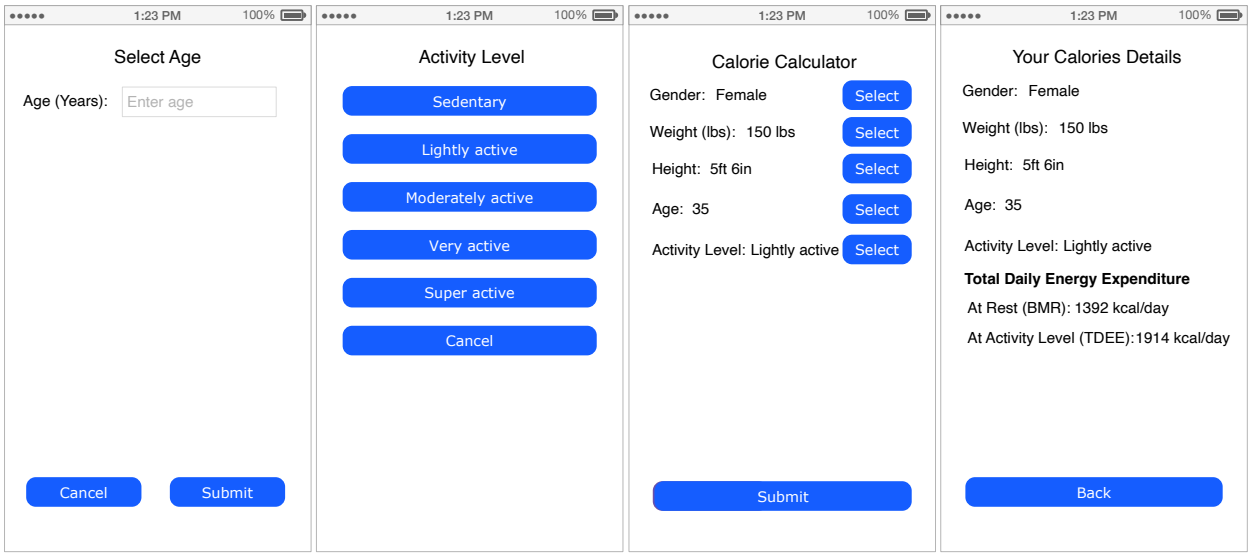
This UIViewController is shown in Figure 1(a). The requirements are listed below:

1. The UIViewController requests the gender, weight, height, age and activity level.
2. Clicking on the "Select" should:
   a. Segue to the corresponding Select UIViewController. Upon returning from the Select UIViewController, the selected value should be displayed in the UILabel as shown in Figure 2(c).
3. Clicking on the Submit button should:
   a. If any of the entries is not selected then show an alert dialog message indicating that the corresponding input is required.
   b. If all the entries are selected then create a Calorie object which contains all the selected values. Segue to the Calorie Details UIViewController and send it the Calorie object.

Part 2, Select Gender, Weight, Height, Age, Activity Level (50 Points):

These UIViewControllers allow the user to select a value, which will be sent to the Calorie Calculator fragment. The requirements are listed below:

1. Clicking the "Submit" button should:
   a. If entry not entered or no selection, then show an alert dialog message indicating that the selection/entry is required.
   b. If entry or selection is provided then send back the selected/entered value to the Calorie Calculator UIViewController using your choice of Unwind Segue, Protocol with Delegate or UINotification approach.
   c. Dismiss this UIViewController.
2. Clicking "Cancel" should dismiss this UIViewController.



(a) Select Age **(b) Select Activity Level** **(c) Calorie Calculator** **(d) Calories Details**

Figure 2, Application User Interface

Part 3, Calorie Details UIViewController (30 Points):

This UIViewController displays the selected calorie details as shown in Figure 2(d). The requirements are listed below:

1. This UIViewController should receive a Calorie object from the Calorie Calculator UIViewController.
2. Display the values as shown in Figure 2(d).
3. Calculate and display the following:
   a. Calories at Rest (BMR).
   b. Calories at Activity Level (TDEE)
4. Clicking “Back” should dismiss this UIViewController.

Calculating Daily Calories:

- This requires two steps, first calculate the Basal Metabolic Rate (BMR), then calculate the total daily energy expenditure (TDEE).

Step 1: Calculating the BMR which represents the number of calories your body needs to maintain its current weight without any additional activity.

$$\text{BMR Men} = \left(\frac{10 \times \text{lbs}}{2.205}\right) + \left(15.875 \times ((12 \times \text{ft}) + \text{in})\right) - \left(5 \times \text{age}\right) + 5$$

$$\text{BMR Women} = \left(\frac{10 \times \text{lbs}}{2.205}\right) + \left(15.875 \times ((12 \times \text{ft}) + \text{in})\right) - \left(5 \times \text{age}\right) - 161$$

Variable	Description
lbs	Weight in pounds
ft and in	Height feet and inches
Age	Age in years

Table 1: BAC variable description

Step 2: Once you have your BMR, adjust it based on your activity level to get your Total Daily Energy Expenditure (TDEE):

Activity Level	Calculate TDEE
Sedentary (little to no exercise)	$1.2 \times$ BMR
Lightly active (light exercise/sports 1-3 days/week)	$1.375 \times$ BMR
Moderately active (moderate exercise 3-5 days/week)	$1.55 \times$ BMR
Very active (hard exercise/sports 6-7 days a week)	$1.725 \times$ BMR
Super active (very hard exercise & physical job or 2x training)	$1.9 \times$ BMR

Examples:

Gender	Weight	Height	Age	Level	BMR	TDEE
Female	150	5ft 6in	35	Lightly active	1392	1914
Male	200	6ft 2in	40	Lightly active	1866	2594
Female	180	5ft 6in	35	Very active	1528	2635
Male	220	6ft 2in	40	Very active	1977	3411

Section:	
Student Name:	
Student ID:	

Part #	Features	Total	Grade	Comments
Part 1	Transitions to all the select screens as indicated in Part 1.	10		
Part 2	Select Gender, sends back the gender value to the Calories Calculator screen.	10		
Part 2	Select Weight, sends back the weight value to the Calories Calculator screen.	10		
Part 2	Select Height, sends back the height value to the Calories Calculator screen.	10		
Part 2	Select Age, sends back the age value to the Calories Calculator screen.	10		
Part 2	Select Activity Level, sends back the activity level value to the Calories Calculator screen.	10		
Part 1	Input validation, transition to Calorie Details, and sends it the Calorie object.	10		
Part 3	Calorie Details, receives the Calorie object, calculates the BMR, and TDEE. Displays the values as shown in Part 3.	30		
	Total	**100**		

Table 1: Grading Key